

Dataset Analysis for Decision-Making

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HIV_Epidemiology... .csv
19.2 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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Key Characteristics of the Dataset

The dataset contains information about HIV/AIDS-related deaths and mother-to-child transmission rates for children and adolescents across different regions, countries, sexes, and age groups. Here are its key characteristics:

- 1. **Geographical Scope :**
 - Data is organized by country (e.g., Sudan, South Sudan, Qatar) and region (e.g., Eastern and Southern Africa, East Asia and Pacific, Middle East and North Africa).
- 2. **Metrics :**
 - Two main metrics are provided:
 - **Estimated number of annual AIDS-related deaths :** Absolute counts of deaths.
 - **Estimated rate of annual AIDS-related deaths :** Deaths per 100,000 population.
 - Additional data includes **mother-to-child transmission rates** as a percentage.
- 3. **Demographic Breakdown :**
 - **Sex :** Both, Female (F), Male (M).
 - **Age Groups :** 0-4, 5-9, 10-14, 10-19, 15-19, etc.
- 4. **Temporal Coverage :**
 - Data spans multiple years, from 2005 to 2023.
- 5. **Confidence Intervals :**
 - For most metrics, lower and upper bounds of estimates are provided.
- 6. **Granularity :**
 - Data is highly granular, allowing analysis at the level of specific age groups, sexes, and regions.

Analytical Question

Question : What was the estimated number of annual AIDS-related deaths for children aged 0-14 in South Sudan in 2015, and how does it compare between males and females?

Answer

In 2015, the estimated number of annual AIDS-related deaths for children aged 0-14 in South Sudan was:

- **Both sexes combined :** 2,000 (lower bound: 1,500, upper bound: 2,500).
- **Females :** 870 (lower bound: 660, upper bound: 1,100).
- **Males :** 1,100 (lower bound: 840, upper bound: 1,400).

Males had a slightly higher estimated number of deaths compared to females.

Python Code

Below is the Python code used to extract and process the data:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 data = pd.read_csv("HIV_Epidemiology_Children_Adolescents_2024_out.csv", header=None)
5
6 # Rename columns for clarity
7 data.columns = [
8     "Country", "Region", "Metric", "Year", "Sex", "Age_Group",
9     "Estimate", "Lower_Bound", "Upper_Bound"
10 ]
11
12 # Filter for South Sudan, 2015, age group 0-14
13 filtered_data = data[
14     (data["Country"] == "South Sudan") &
15     (data["Year"] == 2015.0) &
16     (data["Age_Group"] == "0-14") &
17     (data["Metric"] == "Estimated number of annual AIDS-related deaths")
18 ]
19
20 # Display results
21 result = filtered_data[["Sex", "Estimate", "Lower_Bound", "Upper_Bound"]]
22 print(result)
```

Explanation of the Code

- 1. **Data Loading :**
 - The dataset is loaded using `pandas.read_csv()` with no headers, as the file lacks column names.
- 2. **Column Renaming :**
 - Columns are renamed for better readability and ease of filtering.



3. Filtering :

- The dataset is filtered for:
 - Country: South Sudan.
 - Year: 2015.
 - Age Group: 0-14.
 - Metric: Estimated number of annual AIDS-related deaths.

4. Result Extraction :

- Relevant columns (Sex , Estimate , Lower_Bound , Upper_Bound) are selected and displayed.

Output of the Code

The output will look like this:

	Sex	Estimate	Lower_Bound	Upper_Bound
1	0	Both	2000	1500
2	0	Both	2000	2500
3	1	F	870	660
4	2	M	1100	840
				1400

This confirms the answer provided above.

Develop a compartmental SEIR model to estimate vertical HIV transmission rates in 2030 under three scenarios (optimistic, realistic, pessimistic), adjusting for regional variations in case incidence and prevention program coverage.

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Developing a Compartmental SEIR Model for Vertical HIV Transmission in 2030

To develop a compartmental **SEIR (Susceptible-Exposed-Infectious-Recovered)** model to estimate vertical HIV transmission rates in 2030, we will break the problem into several steps:

Step 1: Define the Compartments

The SEIR model is adapted to account for vertical HIV transmission. The compartments are as follows:

1. **S (Susceptible)** : Pregnant individuals who are HIV-negative.
2. **E (Exposed)** : Pregnant individuals who have been exposed to HIV but are not yet infectious.
3. **I (Infectious)** : Pregnant individuals who are HIV-positive and can transmit the virus vertically to their child.
4. **R (Recovered/Prevented)** : Pregnant individuals who are HIV-positive but have access to effective prevention of mother-to-child transmission (PMTCT) programs, making them unable to transmit the virus.

Additionally, we include a compartment for **children** :

- **C (Child Infected)** : Children born with HIV due to vertical transmission.

Step 2: Incorporate Regional Variations

The dataset shows that HIV incidence rates, mother-to-child transmission rates, and PMTCT coverage vary significantly by region. To account for this, we divide the global population into regions based on UNICEF classifications (e.g., Sub-Saharan Africa, South Asia, etc.). Each region will have its own parameters for:

- HIV incidence rate among pregnant women.
- Mother-to-child transmission rate (%).
- PMTCT program coverage (%).

Step 3: Define Scenarios

We define three scenarios for 2030:

1. **Optimistic Scenario** :
 - High PMTCT coverage (e.g., 95% globally).
 - Reduced HIV incidence rates due to improved healthcare and education.
2. **Realistic Scenario** :
 - Moderate PMTCT coverage (e.g., 75% globally).
 - HIV incidence rates remain consistent with current trends.
3. **Pessimistic Scenario** :
 - Low PMTCT coverage (e.g., 50% globally).
 - Increased HIV incidence rates due to reduced funding or worsening conditions.

Step 4: Key Parameters

Using the dataset, we extract key parameters for each region:

- **HIV incidence rate** : Estimated annual new HIV infections per 1,000 uninfected population.
- **Mother-to-child transmission rate** : Percentage of HIV-positive mothers who transmit the virus to their children without intervention.
- **PMTCT effectiveness** : Percentage reduction in vertical transmission due to PMTCT programs.

For simplicity, we assume:

- PMTCT reduces vertical transmission by 80% when fully implemented.
- Population growth rates remain constant across regions.

Step 5: Model Equations

The SEIR model equations are adapted as follows:

- ... -

Susceptible (S):

$$\frac{dS}{dt} = \text{Births} - (\beta \cdot S) - \mu \cdot S$$

Where:

- β = HIV incidence rate (per year).
- μ = Natural mortality rate.

Exposed (E):

$$\frac{dE}{dt} = (\beta \cdot S) - (\sigma \cdot E) - \mu \cdot E$$

Where:

- σ = Rate at which exposed individuals become infectious.

Infectious (I):

$$\frac{dI}{dt} = (\sigma \cdot E) - (\gamma \cdot I) - \mu \cdot I$$

Where:

- γ = Rate at which infectious individuals enter the "Recovered" compartment due to PMTCT.

Recovered/Prevented (R):

$$\frac{dR}{dt} = (\gamma \cdot I) - \mu \cdot R$$

Child Infected (C):

$$\frac{dC}{dt} = (\alpha \cdot I) - (\delta \cdot C)$$

Where:

- α = Vertical transmission rate (adjusted for PMTCT coverage).
- δ = Mortality rate for HIV-infected children.

Step 6: Simulation for 2030

Using Python, we simulate the model for each scenario and region. Below is the Python code:

```
python
1 import numpy as np
2 from scipy.integrate import odeint
3 import pandas as pd
4
5 # Load regional data from the dataset
6 data = pd.read_csv("HIV_Epidemiology_Children_Adolescents_2024_out.csv", header=None)
7
8 # Rename columns for clarity
9 data.columns = [
10     "Country", "Region", "Metric", "Year", "Sex", "Age_Group",
11     "Estimate", "Lower_Bound", "Upper_Bound"
12 ]
13
14 # Extract key parameters for each region
15 def get_parameters(region, scenario):
16     # Example: HIV incidence rate (per 1,000 uninfected population)
17     hiv_incidence_rate = data[
18         (data["Region"] == region) &
19         (data["Metric"] == "Estimated incidence rate (new HIV infection per 1,000 uninfected population)") &
20         (data["Year"] == 2023.0)
21     ][["Estimate"]].values[0] / 1000
22
23     # Mother-to-child transmission rate (%)
24     mtct_rate = data[
25         (data["Region"] == region) &
26         (data["Metric"] == "Estimated mother-to-child transmission rate (%)") &
27         (data["Year"] == 2023.0)
28     ][["Estimate"]].values[0] / 100
29
30     # PMTCT coverage (%)
31     if scenario == "optimistic":
32         pmtct_coverage = 0.95
33     elif scenario == "realistic":
34         pmtct_coverage = 0.75
35     else: # pessimistic
36         pmtct_coverage = 0.50
37
38     # Adjusted vertical transmission rate
39     adjusted_mtct_rate = mtct_rate * (1 - pmtct_coverage * 0.8) # 80% effectiveness
40
41     return hiv_incidence_rate, adjusted_mtct_rate
42
43 # SEIR model equations
44 def seir_model(y, t, beta, sigma, gamma, alpha, delta, mu):
45     S, E, I, R, C = y
46     dS_dt = -beta * S - mu * S
47     dE_dt = beta * S - sigma * E - mu * E
48     dI_dt = sigma * E - gamma * I - mu * I
49     dR_dt = gamma * I - mu * R
50     dC_dt = alpha * I - delta * C
51     return [dS_dt, dE_dt, dI_dt, dR_dt, dC_dt]
52
53 # Simulate for a specific region and scenario
54 def simulate(region, scenario, years=10):
55     # Initial conditions
56     N = 1e6 # Total population
57     S0 = N - 100 # Susceptible
58     E0 = 100 # Exposed
59     I0 = 0 # Infectious
60     R0 = 0 # Recovered/Prevented
61     C0 = 0 # Child Infected
62     y0 = [S0, E0, I0, R0, C0]
63
64     # Time vector
65     t = np.linspace(0, years, years * 12) # Monthly intervals
66
67     # Parameters
```

```

68     beta, alpha = get_parameters(region, scenario)
69     sigma = 1 / 6 # Average incubation period of 6 months
70     gamma = 1 / 12 # Average time to recover/prevent (1 year)
71     delta = 0.1 # Mortality rate for infected children
72     mu = 0.01 # Natural mortality rate
73
74     # Solve ODEs
75     solution = odeint(seir_model, y0, t, args=(beta, sigma, gamma, alpha, delta, mu))
76     S, E, I, R, C = solution.T
77
78     return t, C[-1] # Return final number of infected children
79
80 # Example: Simulate for Sub-Saharan Africa under all scenarios
81 regions = ["Sub-Saharan Africa"]
82 scenarios = ["optimistic", "realistic", "pessimistic"]
83
84 results = {}
85 for region in regions:
86     results[region] = {}
87     for scenario in scenarios:
88         t, infected_children = simulate(region, scenario)
89         results[region][scenario] = infected_children
90
91 print(results)

```



Step 7: Interpret Results

The output will provide the estimated number of vertically transmitted HIV cases in 2030 for each scenario and region.

